Summary: *Stress Prediction Using Wearable Devices*

Stress prediction through wearable technology has evolved into a major research priority for health monitoring. According to recent reviews and empirical studies, modern wearables track multiple physiological signals — such as Heart Rate Variability (HRV), Electrodermal Activity (EDA), respiration rate, and skin temperature — to assess an individual's stress level in real time. These signals are particularly sensitive to autonomic nervous system responses, making them reliable indicators of stress-related physiological changes.

Datasets like WESAD, PhysioNet, and HRV-Sleep have emerged as benchmarks for developing machine learning models such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and deep learning approaches like Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM). Reported accuracies for stress detection models typically range from 85% to 99%, depending on dataset conditions and preprocessing strategies. Quantitative results consistently show that HRV and EDA yield the best signal combinations for distinguishing between stress and non-stress states.

Modern pipelines for wearable-based stress monitoring include stages for data preprocessing, feature extraction, training, and real-time inference. These models can operate both offline (detecting stress retrospectively) and online (real-time intervention). Newer studies highlight the importance of reducing energy consumption in wearable inference systems while maintaining accuracy — an essential step for scalable healthcare deployment.

In practical terms, these systems aim to enable early detection, biofeedback, and mental health monitoring. The integration of real-time dashboards, similar to your project's visualization, enhances interpretability for users. Future work focuses on expanding model generalization across individuals, ensuring data privacy, and incorporating contextual factors (like environment or activity) for richer stress insights